L' Hospital Rule

L'Hospital's Rule Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \to a} f(x) = 0 \quad \text{and} \quad \lim_{x \to a} g(x) = 0$$

or that

$$\lim_{x \to a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \to a} g(x) = \pm\infty$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or ∞/∞.) Then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

EXAMPLE 1 Find $\lim_{x \to 1} \dfrac{\ln x}{x - 1}$.

SOLUTION Since

$$\lim_{x \to 1} \ln x = \ln 1 = 0 \quad \text{and} \quad \lim_{x \to 1} (x - 1) = 0$$

the limit is an indeterminate form of type $\frac{0}{0}$, so we can apply l'Hospital's Rule:

$$\lim_{x \to 1} \frac{\ln x}{x - 1} = \lim_{x \to 1} \frac{\dfrac{d}{dx}(\ln x)}{\dfrac{d}{dx}(x - 1)} = \lim_{x \to 1} \frac{1/x}{1}$$

$$= \lim_{x \to 1} \frac{1}{x} = 1$$

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EXAMPLE 2 Calculate $\lim_{x \to \infty} \dfrac{e^x}{x^2}$.

SOLUTION We have $\lim_{x \to \infty} e^x = \infty$ and $\lim_{x \to \infty} x^2 = \infty$, so the limit is an indeterminate form of type ∞/∞, and l'Hospital's Rule gives

$$\lim_{x \to \infty} \frac{e^x}{x^2} = \lim_{x \to \infty} \frac{\dfrac{d}{dx}(e^x)}{\dfrac{d}{dx}(x^2)} = \lim_{x \to \infty} \frac{e^x}{2x}$$

Since $e^x \to \infty$ and $2x \to \infty$ as $x \to \infty$, the limit on the right side is also indeterminate, but

a second application of l'Hospital's Rule gives

$$\lim_{x \to \infty} \frac{e^x}{x^2} = \lim_{x \to \infty} \frac{e^x}{2x} = \lim_{x \to \infty} \frac{e^x}{2} = \infty$$

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EXAMPLE 3 Calculate $\lim_{x\to\infty} \frac{\ln x}{\sqrt[3]{x}}$.

SOLUTION Since $\ln x \to \infty$ and $\sqrt[3]{x} \to \infty$ as $x \to \infty$, l'Hospital's Rule applies:

$$\lim_{x\to\infty} \frac{\ln x}{\sqrt[3]{x}} = \lim_{x\to\infty} \frac{1/x}{\frac{1}{3}x^{-2/3}}$$

Notice that the limit on the right side is now indeterminate of type $\frac{0}{0}$. But instead of applying l'Hospital's Rule a second time as we did in Example 2, we simplify the expression and see that a second application is unnecessary:

$$\lim_{x\to\infty} \frac{\ln x}{\sqrt[3]{x}} = \lim_{x\to\infty} \frac{1/x}{\frac{1}{3}x^{-2/3}} = \lim_{x\to\infty} \frac{3}{\sqrt[3]{x}} = 0$$

EXAMPLE 4 Find $\lim_{x\to 0} \frac{\tan x - x}{x^3}$. (See Exercise 1.5.44.)

SOLUTION Noting that both $\tan x - x \to 0$ and $x^3 \to 0$ as $x \to 0$, we use l'Hospital's Rule:

$$\lim_{x\to 0} \frac{\tan x - x}{x^3} = \lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2}$$

Since the limit on the right side is still indeterminate of type $\frac{0}{0}$, we apply l'Hospital's Rule again:

$$\lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2} = \lim_{x\to 0} \frac{2\sec^2 x \tan x}{6x}$$

Because $\lim_{x\to 0} \sec^2 x = 1$, we simplify the calculation by writing

$$\lim_{x\to 0} \frac{2\sec^2 x \tan x}{6x} = \frac{1}{3} \lim_{x\to 0} \sec^2 x \cdot \lim_{x\to 0} \frac{\tan x}{x} = \frac{1}{3} \lim_{x\to 0} \frac{\tan x}{x}$$

We can evaluate this last limit either by using l'Hospital's Rule a third time or by writing $\tan x$ as $(\sin x)/(\cos x)$ and making use of our knowledge of trigonometric limits. Putting together all the steps, we get

$$\begin{aligned}\lim_{x\to 0} \frac{\tan x - x}{x^3} &= \lim_{x\to 0} \frac{\sec^2 x - 1}{3x^2} = \lim_{x\to 0} \frac{2\sec^2 x \tan x}{6x} \\ &= \frac{1}{3} \lim_{x\to 0} \frac{\tan x}{x} = \frac{1}{3} \lim_{x\to 0} \frac{\sec^2 x}{1} = \frac{1}{3}\end{aligned}$$

Practice Questions

8–68 Find the limit. Use l'Hospital's Rule where appropriate. If there is a more elementary method, consider using it. If l'Hospital's Rule doesn't apply, explain why.

8. $\lim_{x\to 3} \frac{x-3}{x^2-9}$

9. $\lim_{x\to 4} \frac{x^2-2x-8}{x-4}$

10. $\lim_{x\to -2} \frac{x^3+8}{x+2}$

11. $\lim_{x\to 1} \frac{x^3 - 2x^2 + 1}{x^3 - 1}$

12. $\lim_{x\to 1/2} \frac{6x^2 + 5x - 4}{4x^2 + 16x - 9}$

13. $\lim_{x\to(\pi/2)^+} \frac{\cos x}{1 - \sin x}$

14. $\lim_{x\to 0} \frac{\tan 3x}{\sin 2x}$

15. $\lim_{t\to 0} \frac{e^{2t} - 1}{\sin t}$

16. $\lim_{x\to 0} \frac{x^2}{1 - \cos x}$

17. $\lim_{\theta\to\pi/2} \frac{1 - \sin\theta}{1 + \cos 2\theta}$

18. $\lim_{\theta\to\pi} \frac{1 + \cos\theta}{1 - \cos\theta}$

19. $\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}}$

20. $\lim_{x\to\infty} \frac{x + x^2}{1 - 2x^2}$

21. $\lim_{x\to 0^+} \frac{\ln x}{x}$

22. $\lim_{x\to\infty} \frac{\ln\sqrt{x}}{x^2}$

23. $\lim_{t\to 1} \frac{t^8 - 1}{t^5 - 1}$

24. $\lim_{t\to 0} \frac{8^t - 5^t}{t}$

25. $\lim_{x\to 0} \frac{\sqrt{1 + 2x} - \sqrt{1 - 4x}}{x}$

26. $\lim_{u\to\infty} \frac{e^{u/10}}{u^3}$

27. $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$

28. $\lim_{x\to 0} \frac{\sinh x - x}{x^3}$

29. $\lim_{x\to 0} \frac{\tanh x}{\tan x}$

30. $\lim_{x\to 0} \frac{x - \sin x}{x - \tan x}$

31. $\lim_{x\to 0} \frac{\sin^{-1}x}{x}$

32. $\lim_{x\to\infty} \frac{(\ln x)^2}{x}$

33. $\lim_{x\to 0} \frac{x3^x}{3^x - 1}$

34. $\lim_{x\to 0} \frac{\cos mx - \cos nx}{x^2}$

35. $\lim_{x\to 0} \frac{\ln(1 + x)}{\cos x + e^x - 1}$

36. $\lim_{x\to 1} \frac{x\sin(x - 1)}{2x^2 - x - 1}$

37. $\lim_{x\to 0^+} \frac{\arctan(2x)}{\ln x}$

38. $\lim_{x\to 0^+} \frac{x^x - 1}{\ln x + x - 1}$

39. $\lim_{x\to 1} \frac{x^a - 1}{x^b - 1}$, $b \neq 0$

40. $\lim_{x\to 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

41. $\lim_{x\to 0} \frac{\cos x - 1 + \frac{1}{2}x^2}{x^4}$

42. $\lim_{x\to a^+} \frac{\cos x \ln(x - a)}{\ln(e^x - e^a)}$

43. $\lim_{x\to\infty} x\sin(\pi/x)$

44. $\lim_{x\to\infty} \sqrt{x}\, e^{-x/2}$

45. $\lim_{x\to 0} \sin 5x \csc 3x$

46. $\lim_{x\to -\infty} x\ln\left(1 - \frac{1}{x}\right)$

47. $\lim_{x\to\infty} x^3 e^{-x^2}$

48. $\lim_{x\to\infty} x^{3/2}\sin(1/x)$

49. $\lim_{x\to 1^+} \ln x \tan(\pi x/2)$

50. $\lim_{x\to(\pi/2)^-} \cos x \sec 5x$

51. $\lim_{x \to 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$

52. $\lim_{x \to 0} (\csc x - \cot x)$

53. $\lim_{x \to 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$

54. $\lim_{x \to 0^+} \left(\frac{1}{x} - \frac{1}{\tan^{-1} x} \right)$

55. $\lim_{x \to \infty} (x - \ln x)$

56. $\lim_{x \to 1^+} [\ln(x^7 - 1) - \ln(x^5 - 1)]$

57. $\lim_{x \to 0^+} x^{\sqrt{x}}$

58. $\lim_{x \to 0^+} (\tan 2x)^x$

59. $\lim_{x \to 0} (1 - 2x)^{1/x}$

60. $\lim_{x \to \infty} \left(1 + \frac{a}{x} \right)^{bx}$

61. $\lim_{x \to 1^+} x^{1/(1-x)}$

62. $\lim_{x \to \infty} x^{(\ln 2)/(1 + \ln x)}$

63. $\lim_{x \to \infty} x^{1/x}$

64. $\lim_{x \to \infty} x^{e^{-x}}$

65. $\lim_{x \to 0^+} (4x + 1)^{\cot x}$

66. $\lim_{x \to 1} (2 - x)^{\tan(\pi x/2)}$

67. $\lim_{x \to 0^+} (1 + \sin 3x)^{1/x}$

68. $\lim_{x \to \infty} \left(\frac{2x - 3}{2x + 5} \right)^{2x+1}$